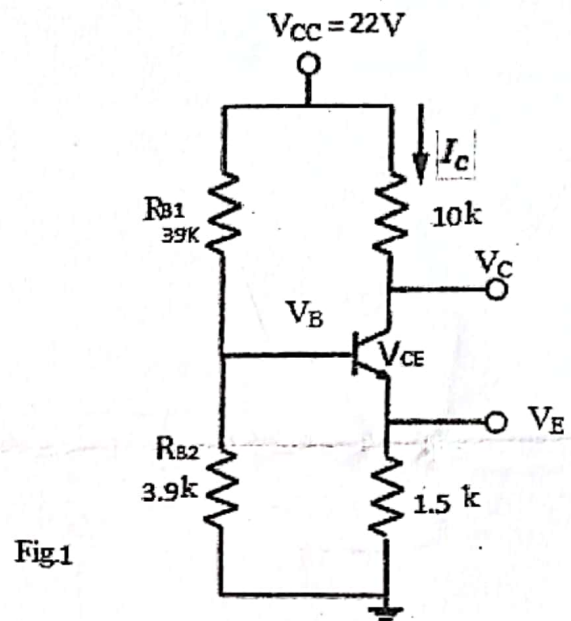


ANSWER ANY FOUR QUESTIONS

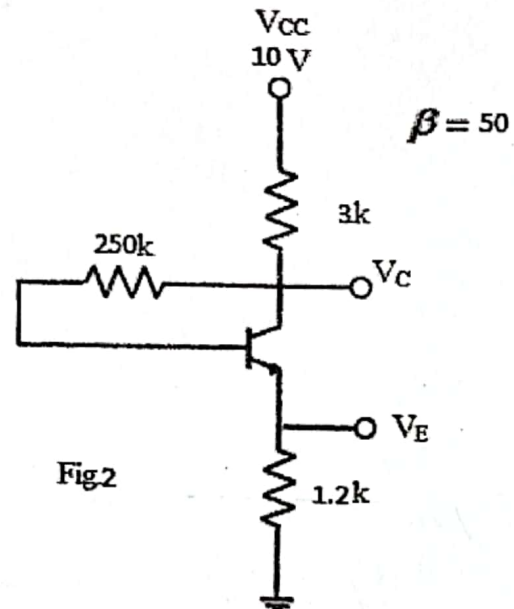
Question 1

Calculate V_{CE} and I_C of Fig.1



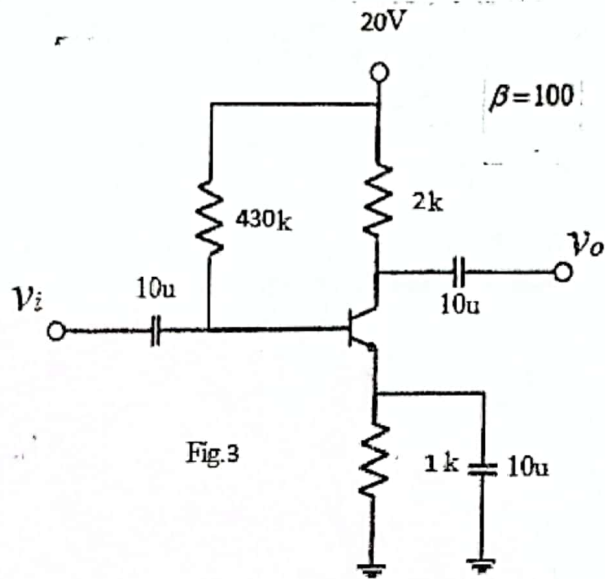
Question 2

Calculate I_E and V_{CE} using voltage feedback



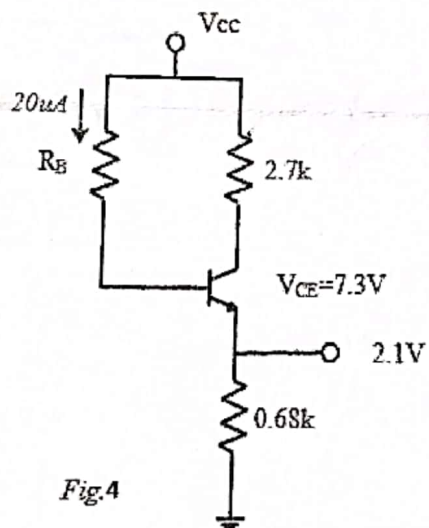
Question 3

Calculate V_{CE} and I_C of Fig.3



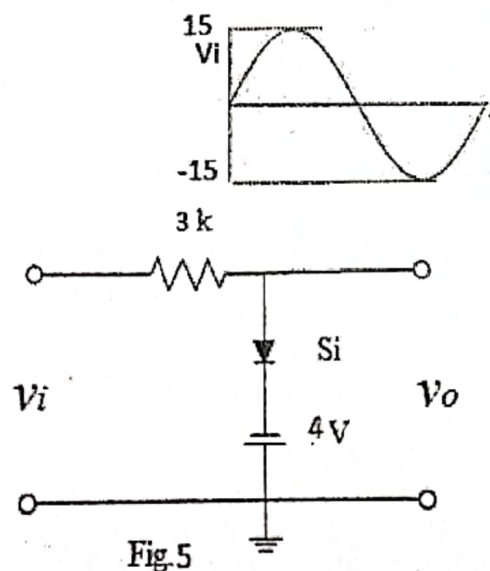
Question 4

Calculate β , V_{CC} , I_C , R_B



Question 5

Sketch and calculate V_o of Fig.5



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odd Semester Examinations Academic Year 2017/2018

Subject: Electronics Circuit

Class: B3CS/B3CSE

Dr. A. Bawazir

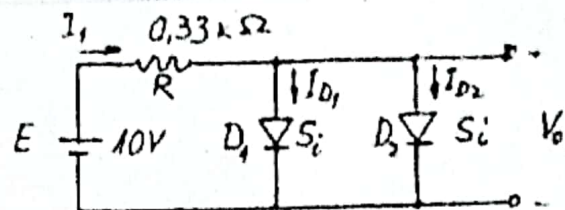
Date 15.01.2018

Time: 2 Hours

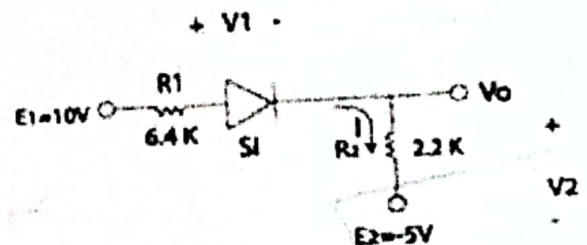
ANSWER ANY FOUR QUESTIONS

Question 1

a) Determine V_o , I_1 , I_{D1} and I_{D2} for the parallel diode configuration of Figures.



b) Determine I , V_1 , V_2 and V_o for the series dc configuration of figure.

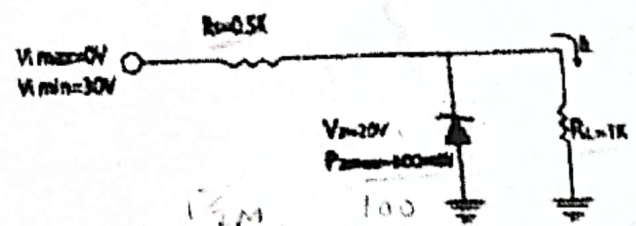


$$E_1 - V_{R1} - V_o - V_{R2} - E_2 = 0$$

$$10 - 6.4 I_1 - 0.7 - 2.2 I_2 + 5 = 0$$

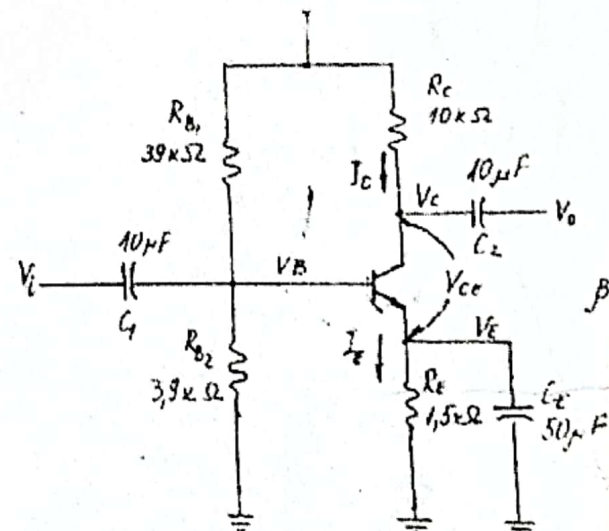
Question 2

Design the voltage regulator that will maintain an output voltage of 20V across a 1-K load with an input that will vary between 30 and 50 V. That is determine proper value of R_s and maximum current I_{zmax}



Question 3

Calculate the dc bias voltage V_{ce} and current I_c for the circuit of Figure



$$I_{D1} = \frac{V_{D1} - V_{D1(sat)}}{R_{D1} + (1+\beta)R_E}$$

$$At V_{in} = 30$$

$$I_{S1} = \frac{V_{i1} - V_{Z1}}{R_{S1}}$$

$$I_{S1} = I_{Z1(max)} + I_{L1(max)}$$

$$I_{L1} = \frac{V_{L1}}{R_{L1}}$$

$$I_C = I_B \cdot \beta$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta) R_E}$$

Question 4

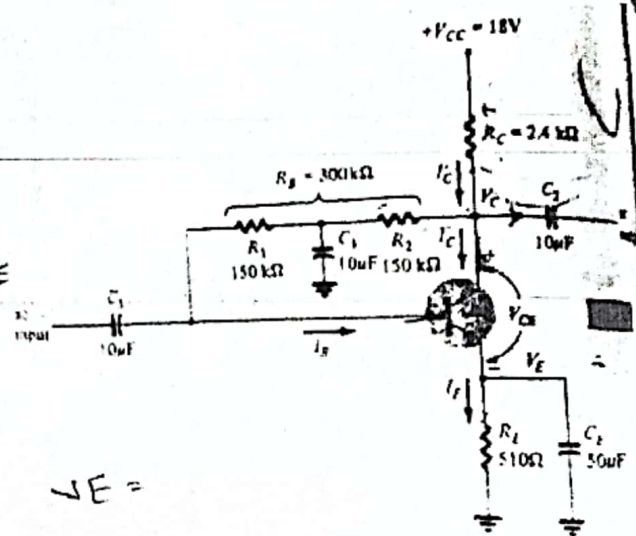
Calculate the dc collector current I_C and voltage V_C for bias circuit of figure

$$V_C = V_{CC} - I_C R_C - I_E R_E$$

$$I_C = I_{BB}$$

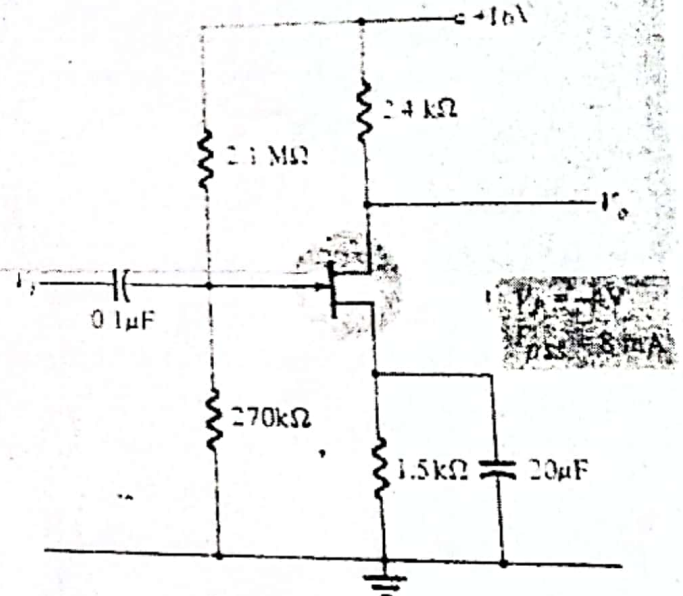
$$I_{BB} = \frac{V_E}{R_E}$$

$$V_E =$$



Question 5

Determine the bias current I_D in the circuit of figure.



Question 6

Calculate the bias values of I_D and V_{DS} for the circuit of figure.

$$V_{CE} = V_C - V_E$$

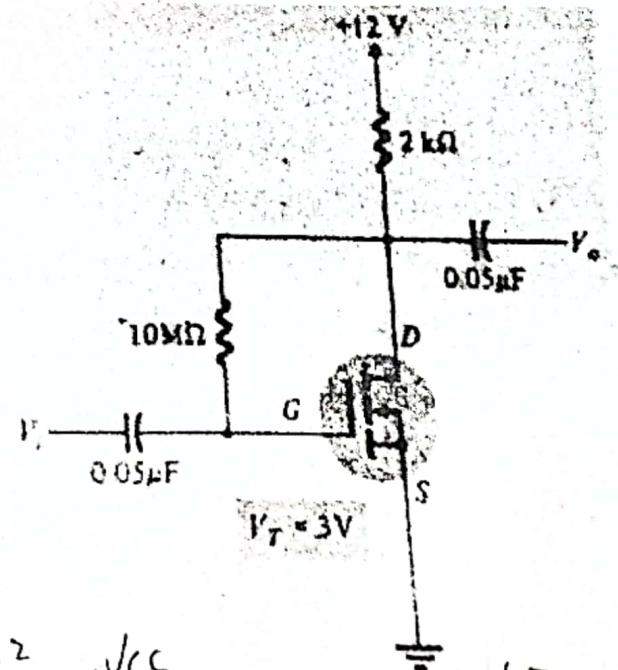
$$\Rightarrow V_C = V_{CC} - I_C R_C$$

$$I_C = \frac{V_E}{R_E}$$

$$V_{CE} = V_{C1} - V_{C2} - V_{C3}$$

$$V_B = \frac{R_{B2}}{R_{B1} + R_{B2}} \cdot V_{CC}$$

42.9



Circuit

①

UNIVERSITY OF ADEN
Department of Electronics and Communication

Even Semester Examinations Academic Year 2017/2018
Subject: Electronics Circuit

Dr. A. Bawazir

Date 14.05.2018

Class: B2CS/B2CSE

Time: 2 Hours

ANSWER ANY FOUR QUESTIONS

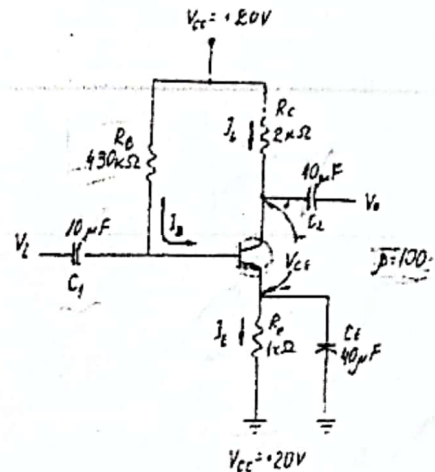
Question 1

Calculate the dc bias voltage V_{CE} and current I_C in the circuit of Figure

$$V_{CE} = V_{CC} - I_C R_C$$

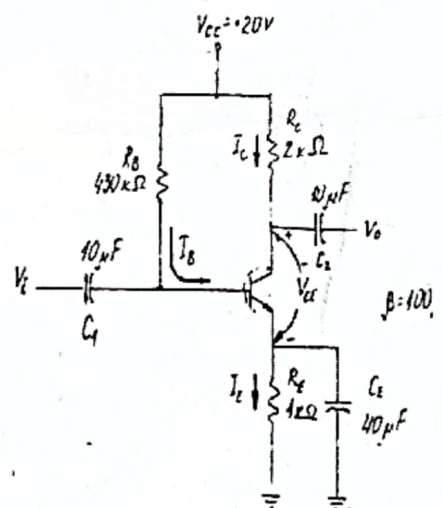
$$V_{CC} = 10V, R_C = 2k\Omega, I_C = 1mA$$

$$V_{CE} = 10V - (1mA)(2k\Omega) = 8V$$



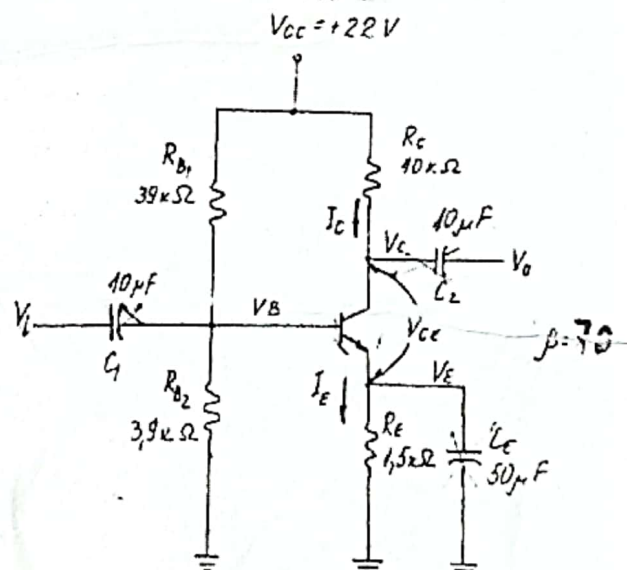
Question 2

Calculate the value of collector resistor R_C needed to obtain $V_C = 10V$, using the circuit of Figure



Question 3

Calculate the dc bias voltage V_{CE} and current I_C for the circuit of Figure



UNIVERSITY OF ADEN

Department of Electronics and Communication

Even Semester Examinations Academic Year 2018/2019

Subject: Electronics Circuit

Dr. A. Bawazir

Date: 8/3/2019

Class: B2CS/E

Time: 2 Hours

ANSWER ANY FOUR QUESTIONS

Question 1

Calculate V_{CE} and I_C of Fig.1

~~Fig.1~~
~~Fig.1~~
~~Fig.1~~

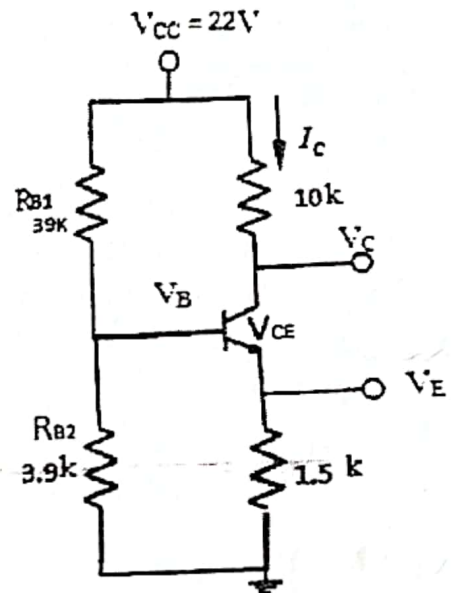


Fig1

Question 2

Calculate I_E and V_{CE} using voltage feedback

$I_E = 1.02$
 $V_{CE} = 5.716$

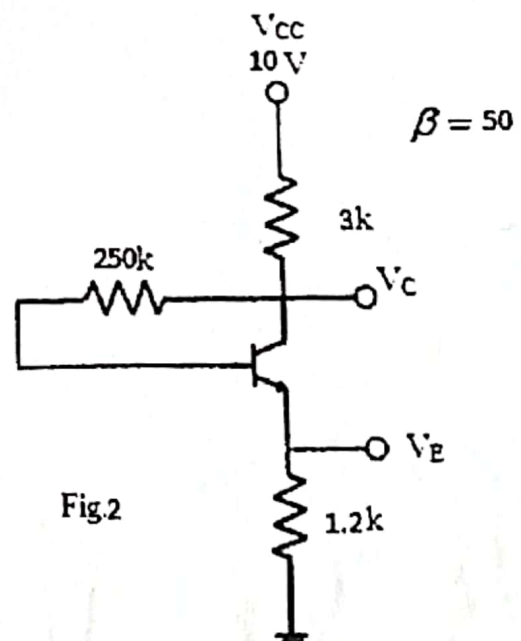
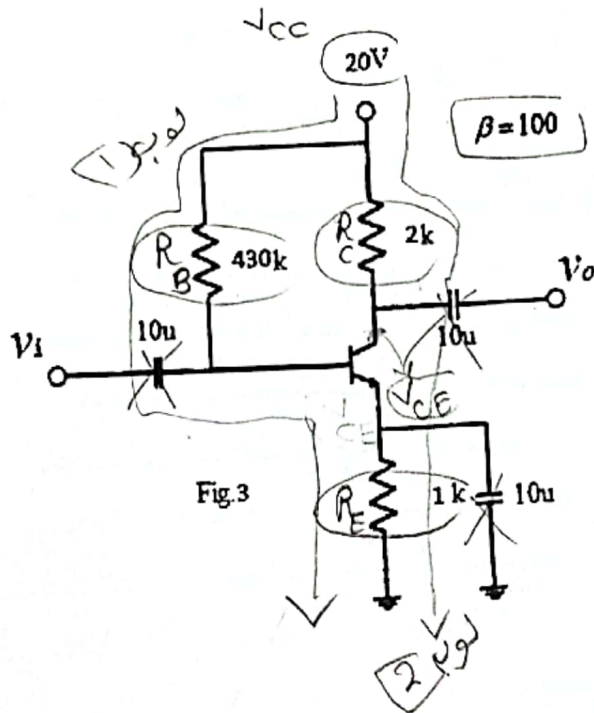


Fig.2

Question 3

Calculate V_{CE} and I_C of Fig.3

$$\begin{aligned} I_C &= 36.24 \mu A \\ &= 3.624 \text{ mA} \\ &= 9.698 \text{ V} \end{aligned}$$

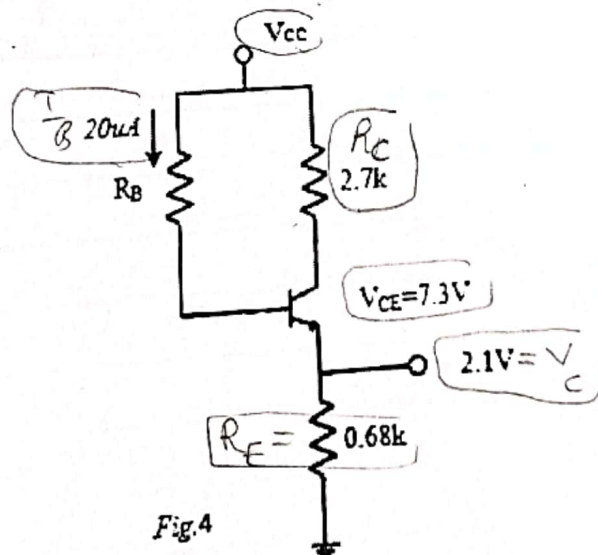


Question 4

Calculate β , V_{CC} , I_C , R_B

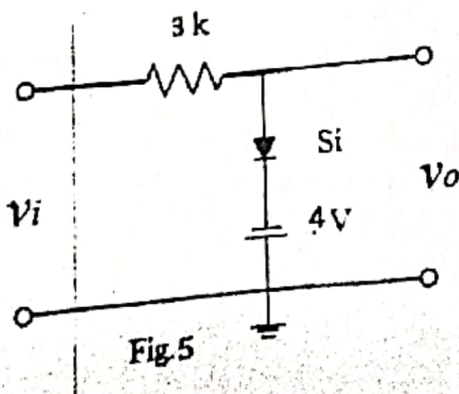
$$\begin{aligned} \beta &= 154 \\ V_{CC} &= 17.71 \text{ V} \\ I_C = I_E &= 2.08 \text{ mA} \\ V_{CE} &= 14.93 \text{ V} \end{aligned}$$

$$\beta = \frac{I_C}{I_B}$$

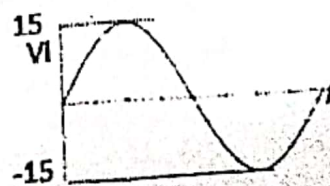


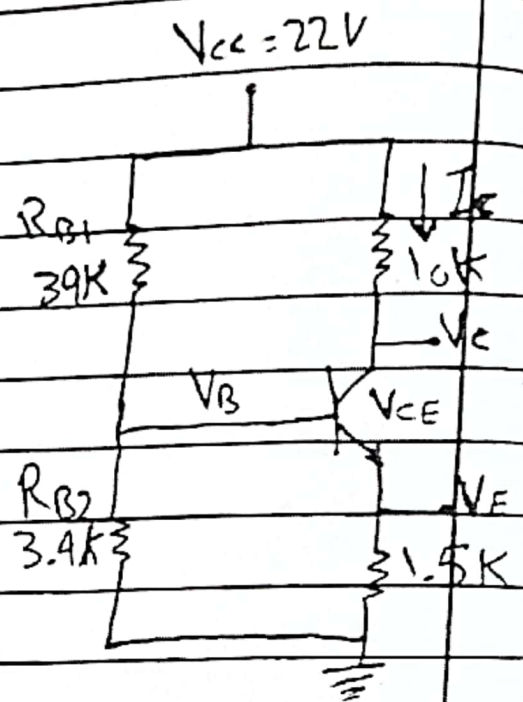
Question 5

Sketch and calculate V_o of Fig.5



$$\begin{aligned} +\frac{1}{2}) V_o &= -4 + 0.7 = -3.3 \text{ V} \\ -\frac{1}{2}) & \end{aligned}$$



Q.1) $V_{CE} = ?$, $I_C = ?$ 

$$V_B = \frac{R_{B2} \cdot V_{CC}}{R_{B1} + R_{B2}} = \frac{3.9 \times 22}{39 + 3.9} = 2 \text{ V}$$

$$V_{BE} = V_B - V_E \Rightarrow V_E = V_B - V_{BE} = 2 - 0.7 = 1.3 \text{ V}$$

$$I_C \approx I_E = \frac{V_E}{R_E} = \frac{1.3}{1.5} = 0.866 \text{ mA}$$

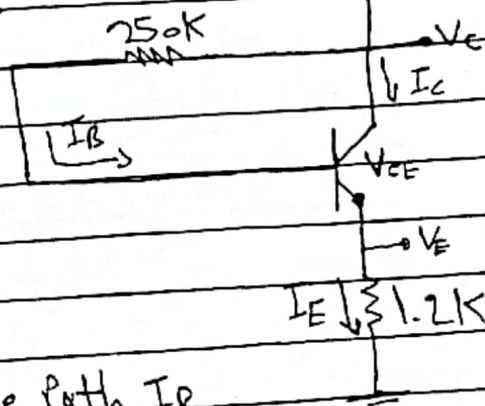
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 22 - 0.866 (10 + 1.5) = 12.04 \text{ V}$$

Q.2) $I_E = ?$, $V_{CE} = ?$

$$\bar{I}_C = I_C + I_B \approx I_C = \beta I_B$$



$\beta = 50$

Apply KVL on the path I_B

$$V_{CC} - \bar{I}_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - \beta I_B R_C - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + R_C + (\beta + 1) R_E} = \frac{10 - 0.7}{50 \times 3 + 250 + (50 + 1) \times 1.2}$$

$$I_B = 20.16 \mu A$$

$$I_C = \beta I_B = 50 \times 20.16 = 1.008 \text{ mA}$$

$$I_E = I_C + I_B = 1.008 + 0.02016 = 1.028 \text{ mA} \approx I_C$$

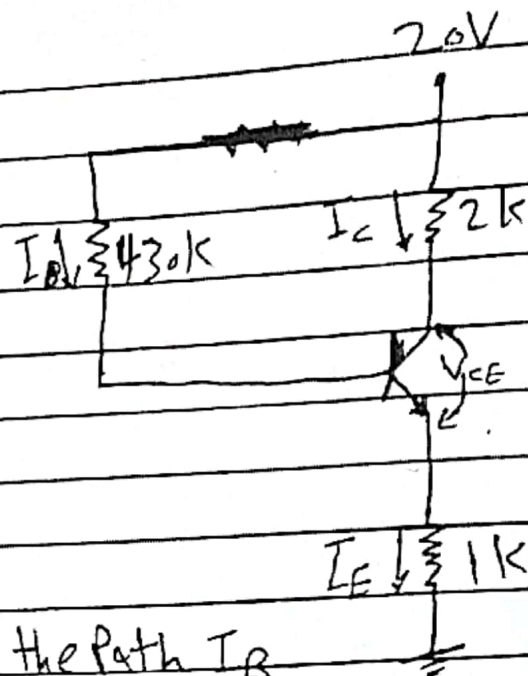
Apply KVL on the path V_{CE}

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 10 - 1.008 (3 + 1.2) = 5.76 \text{ V}$$

Q.3) $V_{CE} = 2$, $I_C = 2$



$\beta = 100$

APPLY KVL act on the Path I_B

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = I_C + I_B = \beta I_B + I_B = (\beta + 1) I_B$$

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} = \frac{20 - 0.7}{430 + (100 + 1) \times 1} = 36.34 \mu A$$

$$I_C = \beta I_B = 100 \times 36.34 = 3.634 \text{ mA}$$

APPLY KVL act on the Path V_{CE}

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_E \approx I_C$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E) = 20 - 3.634 (2 + 1) = 9.098 \text{ V}$$

Q.4) $\beta = 2$, $V_{CC} = 2$, $I_C = 2$, $R_B = 2$

$\beta = 100$

$$I_C = \beta I_B \Rightarrow \beta = \frac{I_C}{I_B}$$

$$I_C \approx I_E = \frac{V_{RE}}{R_E}$$

$$I_E = \frac{2.1}{0.68} = 3.08 \text{ mA} \approx I_C$$

$$\beta = \frac{I_C}{I_B} = \frac{3.08}{20 \times 10^{-5}} = 154$$

APPLY KVL act on the Path V_{CE}

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$I_C \approx I_E$$

$$V_{CC} = V_{CE} + I_E (R_C + R_E) = 7.3 + 3.08(2.7 + 0.68)$$

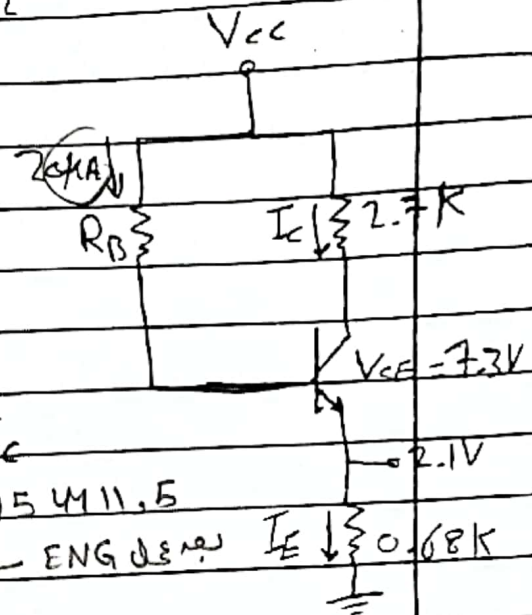
$$V_{CC} = 17.7 \text{ V}$$

APPLY KVL act on the Path I_B

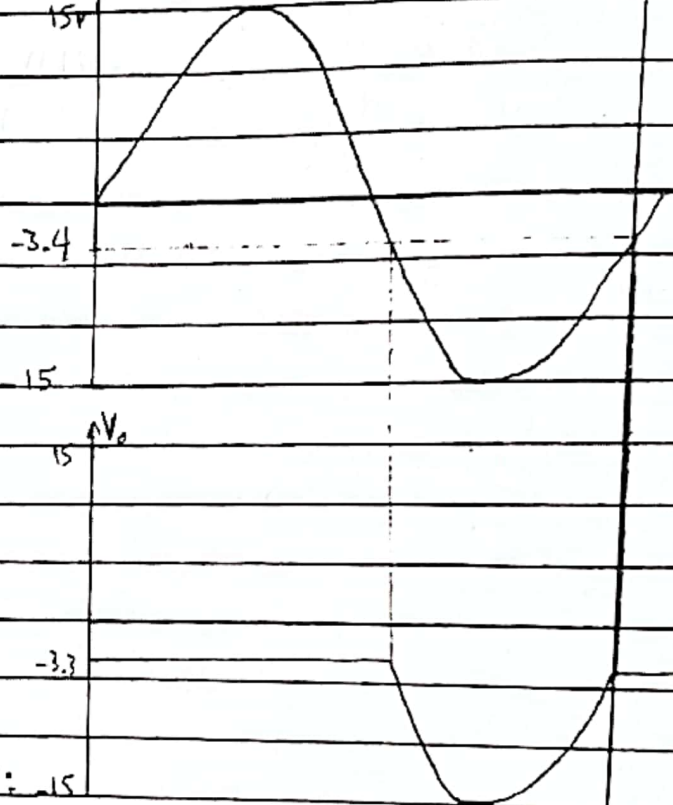
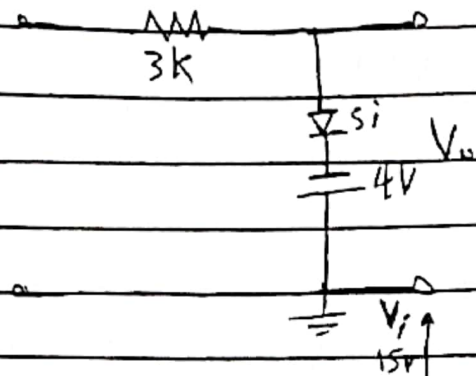
$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$R_B = \frac{V_{CC} - V_{BE} - I_E R_E}{I_B} = \frac{17.71 - 0.7 - 3.08 \times 0.68}{20 \times 10^{-3}}$$

$$R_B = 745.78 \text{ k}\Omega$$



Q.5) Sketch and calculate V_o



* At Positive half cycle:

Diode at on state

$$V_o = -4 + 0.7 = -3.3V$$

* At negative half cycle:

Diode at on state when

$$V_i = 0 \text{ to } -3.3V$$

$$V_o = -4 + 0.7 = -3.3V$$

Diode at off state when

$$V_i = -3.4 \text{ to } -15V$$

$$V_o = V_i$$

Eng. Yassem Mohammed

UNIVERSITY OF ADEN

Department of Electronics and Communication

Even Semester Examinations Academic Year 2017/2018

Dr. A. Bawazir

Subject: Electronics Circuit

Date 14.05.2018

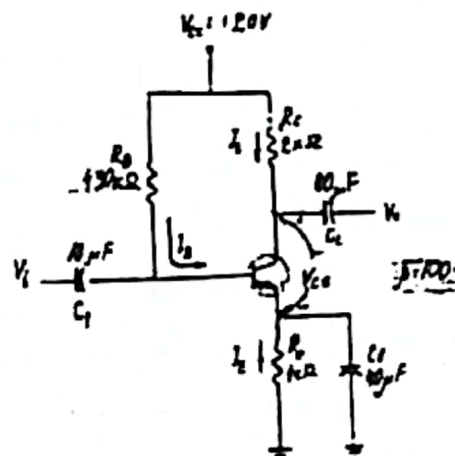
Class: B2CS/B2CSE

Time: 2 Hours

ANSWER ANY FOUR QUESTIONS

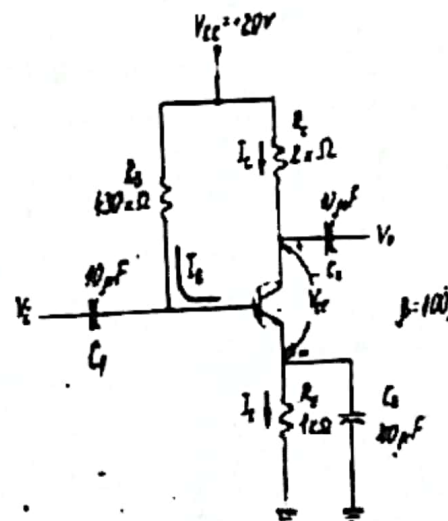
Question 1

Calculate the dc bias voltage (V_{ce}) and current (I_c) in the circuit of Figure



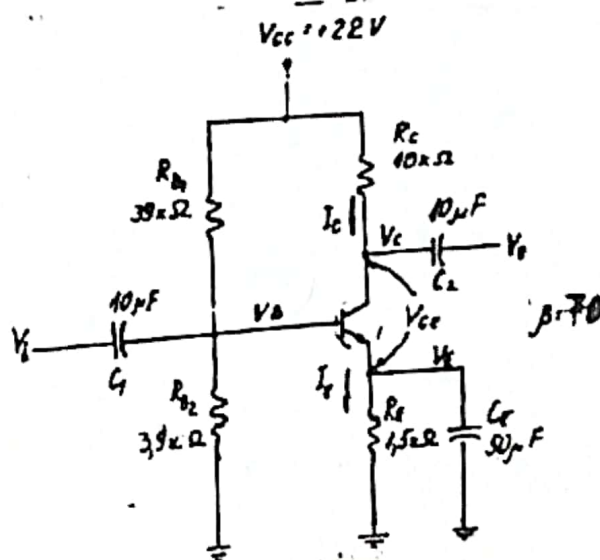
Question 2

Calculate the value of collector resistor (R_c) needed to obtain $V_c = 10V$, using the circuit of Figure



Question 3

Calculate the dc bias voltage (V_{ce}) and current (I_c) for the circuit of Figure

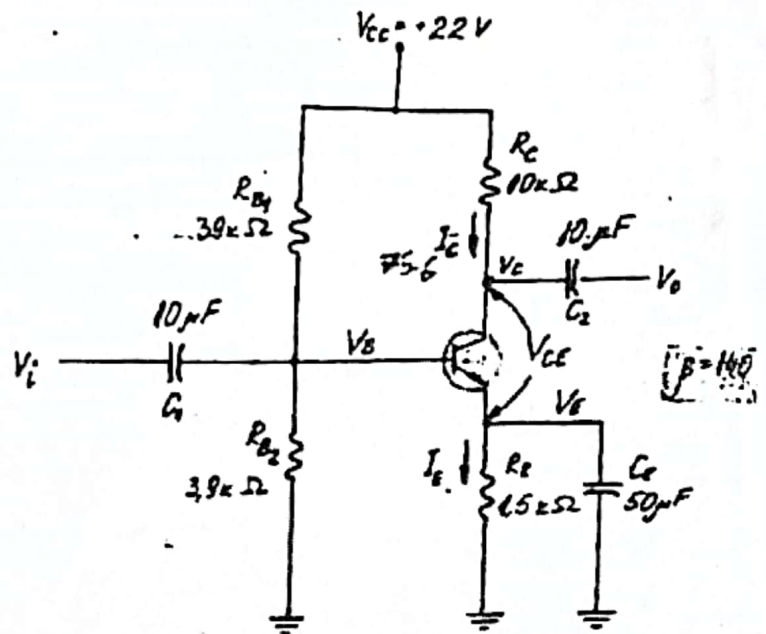


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Question 4

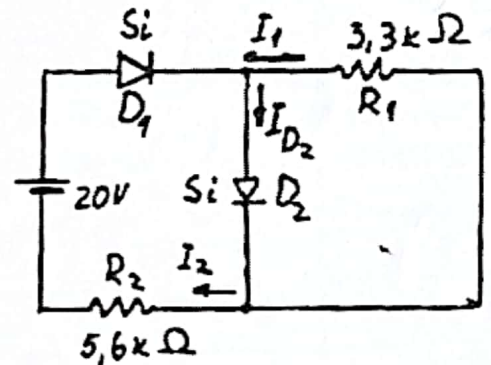
Calculate the dc bias voltage V_{CE} and current I_C for the circuit of Figure



Question 5

Determine the currents I_1 , I_2 and I_{D1} for the network of Figure

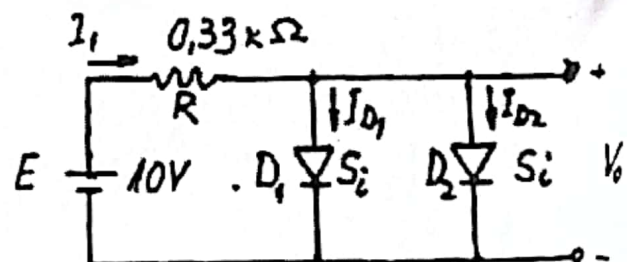
$$\begin{aligned} 20 - V_{D1} - I_1 R_1 - I_2 R_2 &= 0 \\ 20 - V_{D1} - I_1 R_1 - I_2 R_2 &= 0 \\ 20 - V_{D1} - I_1 R_1 - I_2 R_2 &= 0 \end{aligned}$$



Question 6

Determine V_o , I_1 , I_{D1} and I_{D2} for the parallel diode configuration of Figure

$$\begin{aligned} E - I_R R - V_o &= 0 \\ E - V_R - V_{D1} &= 0 \\ V_R &= E - V_{D1} \end{aligned}$$



Q1) $V_{CE} = 2$, $I_C = 3$

$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E \approx I_C$$

$$V_{CC} - I_B R_B - V_{BE} - I_C R_E = 0$$

$$I_C = \beta I_B$$

$$V_{CC} - I_B R_B - V_{BE} - \beta I_B R_E = 0$$

$$V_{CC} - (R_B + \beta R_E) I_B - V_{BE} = 0 \Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_E}$$

$$\therefore I_B = \frac{20 - 0.7}{430 + 100} = 36.41 \mu A$$

$$\therefore I_C = \beta I_B = 100 \times 36.41 = 3.641 \text{ mA}$$

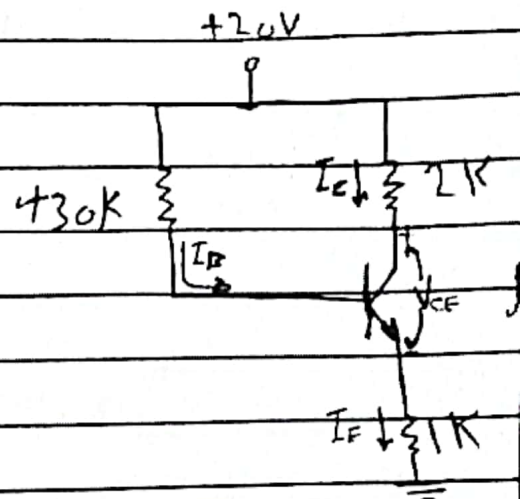
Apply KVL Act on the path of V_{CE}

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CC} - I_C R_C - V_{CE} - \underbrace{(\beta I_B R_E)}_{\text{or } I_C} = 0$$

$$V_{CE} = V_{CC} - I_C R_C - I_C R_E \Rightarrow V_{CE} = V_{CC} - (R_C + R_E) I_C$$

$$\therefore V_{CE} = 20 - (7 + 1) \times 3.641 = 9.077 \text{ V}$$



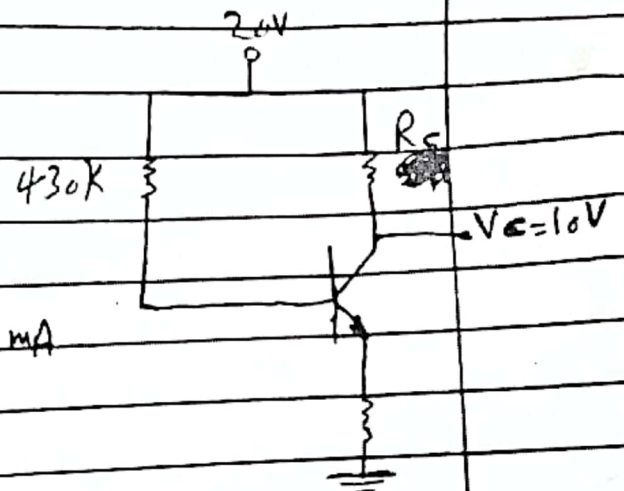
Q2) It's same the Q1, but require Value of R_c for $V_c = 10V$

Applying KVL across the path R_c

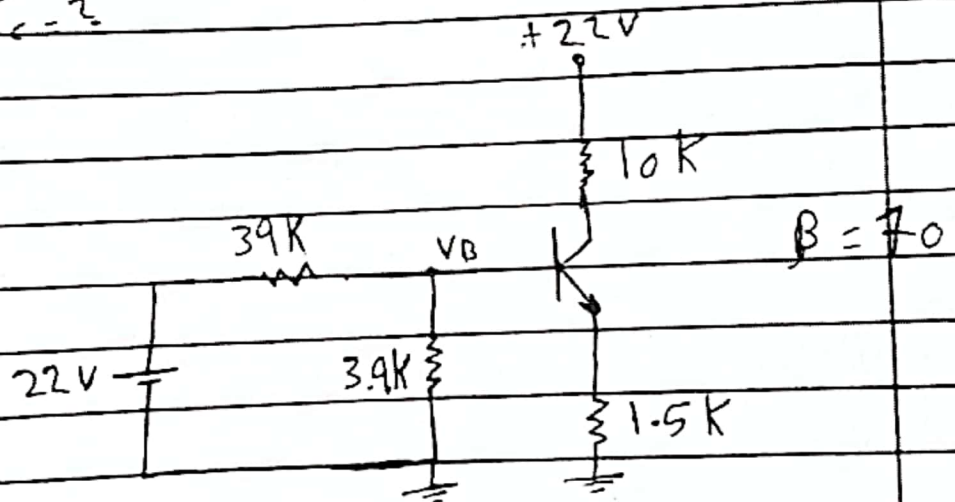
$$V_{cc} - I_c R_c = V_c$$

$$R_c = \frac{V_{cc} - V_c}{I_c}, \text{ At } Q_1, I_c = 3.641 \text{ mA}$$

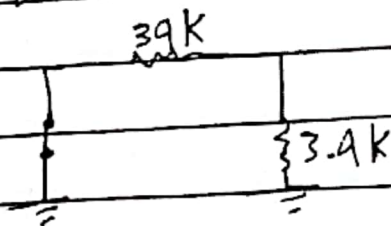
$$R_c = \frac{20 - 10}{3.641} = 2.74 \text{ K}\Omega$$



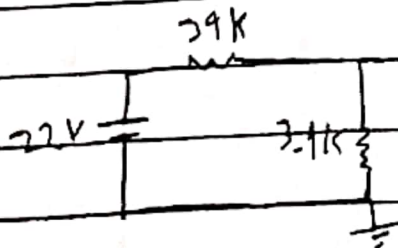
Q3) $V_{CE} = ?$, $I_c = ?$



Applying thevenin's theorem on the input circuit

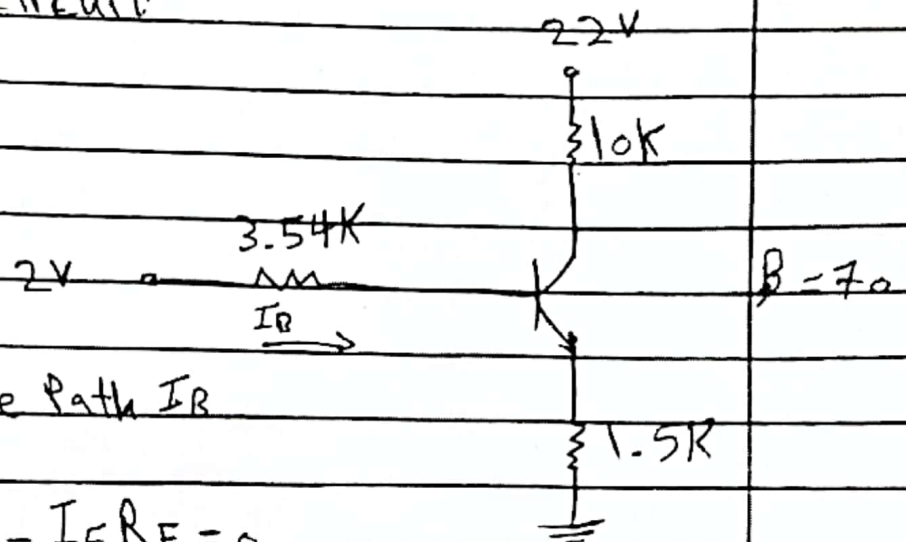


$$R_{th} = R_{B1} || R_{B2} = \frac{39 \times 3.9}{39 + 3.9} = 3.54 \text{ K}\Omega$$



$$V_{th} = V_{R_{B2}} = \frac{R_{B2} V_{cc}}{R_{B1} + R_{B2}} = \frac{3.9 \times 22}{39 + 3.9} = 2 \text{ V}$$

Redrawing the circuit



Apply KVL act on the path I_B

$$V_{th} - I_B R_{th} - V_{BE} - I_E R_E = 0$$

$$V_{th} - I_B R_{th} - V_{BE} - \beta I_B R_E = 0$$

$$V_{th} - (R_{th} + \beta R_E) I_B - V_{BE} = 0 \rightarrow I_B = \frac{V_{th} - V_{BE}}{R_{th} + \beta R_E}$$

$$I_B = \frac{2 - 0.7}{3.54 + 105} = 11.98 \mu A$$

$$I_C = \beta I_B = 70 \times 11.98 = 0.8386 \text{ mA}$$

Now apply KVL act on the path V_{CE}

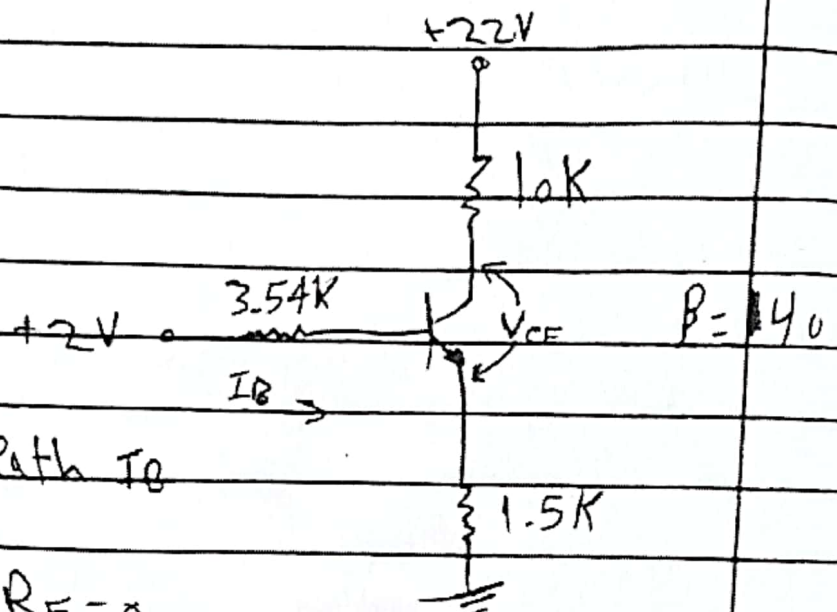
$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0 \rightarrow V_{CC} - (R_C + R_E) I_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - (R_C + R_E) \times I_C$$

$$\therefore V_{CE} = 22 - (10 + 1.5) \times 0.8386 = 12.35 \text{ V}$$

Q4) Same Q3, but $\beta = 140$



APPLY KVL act on the Path T_B

$$V_{TH} - I_B R_{TH} - V_{BE} - I_E R_E = 0$$

$$V_{TH} - (R_{TH} + \beta R_E) I_B - V_{BE} = 0 \Rightarrow I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + \beta R_E}$$

$$I_B = \frac{2 - 0.7}{3.54 + 210} = 6.08 \mu A$$

$$\therefore I_C = \beta I_B = 140 \times 6.08 = 0.8512 \text{ mA}$$

APPLY KVL act on the Path V_{CE}

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CC} - I_C R_C - V_{CE} - I_C R_E = 0 \Rightarrow V_{CE} = V_{CC} - (R_C + R_E) \times I_C$$

$$\therefore V_{CE} = 22 - (10 + 1.5) \times 0.8512 = 12.21 \text{ V}$$

calculate the dc collector current I_C and voltage V_C for bias circuit of figure?

$$R_1 = 100k\Omega, R_2 = 250k\Omega$$

$$= 100 \times 10^3 + 250 \times 10^3$$

$$R_B = 350k\Omega$$

$$V_{CC} - R_1 I_B - R_2 I_B = V_{BE} - R_E I_E$$

$$\Rightarrow V_{CC} - R_B I_B - R_E \beta I_B - V_{BE} - R_E \beta I_B = 0$$

$$V_{CC} - V_{BE} = R_C \beta I_B + R_B I_B + R_E \beta I_B$$

$$\therefore I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{10 - 0.7}{350k + 100(3k + 1.2k)}$$

$$\therefore I_B = 21.95 \mu A$$

$$I_C = \beta I_B = 100 \times 21.95 \times 10^{-6}$$

$$I_C = 2.195 \times 10^{-3} \Rightarrow 2.195 mA$$

الآن نوجد V_C الجزء عند نقطة تقاطع الخطين

بعضية الآخر

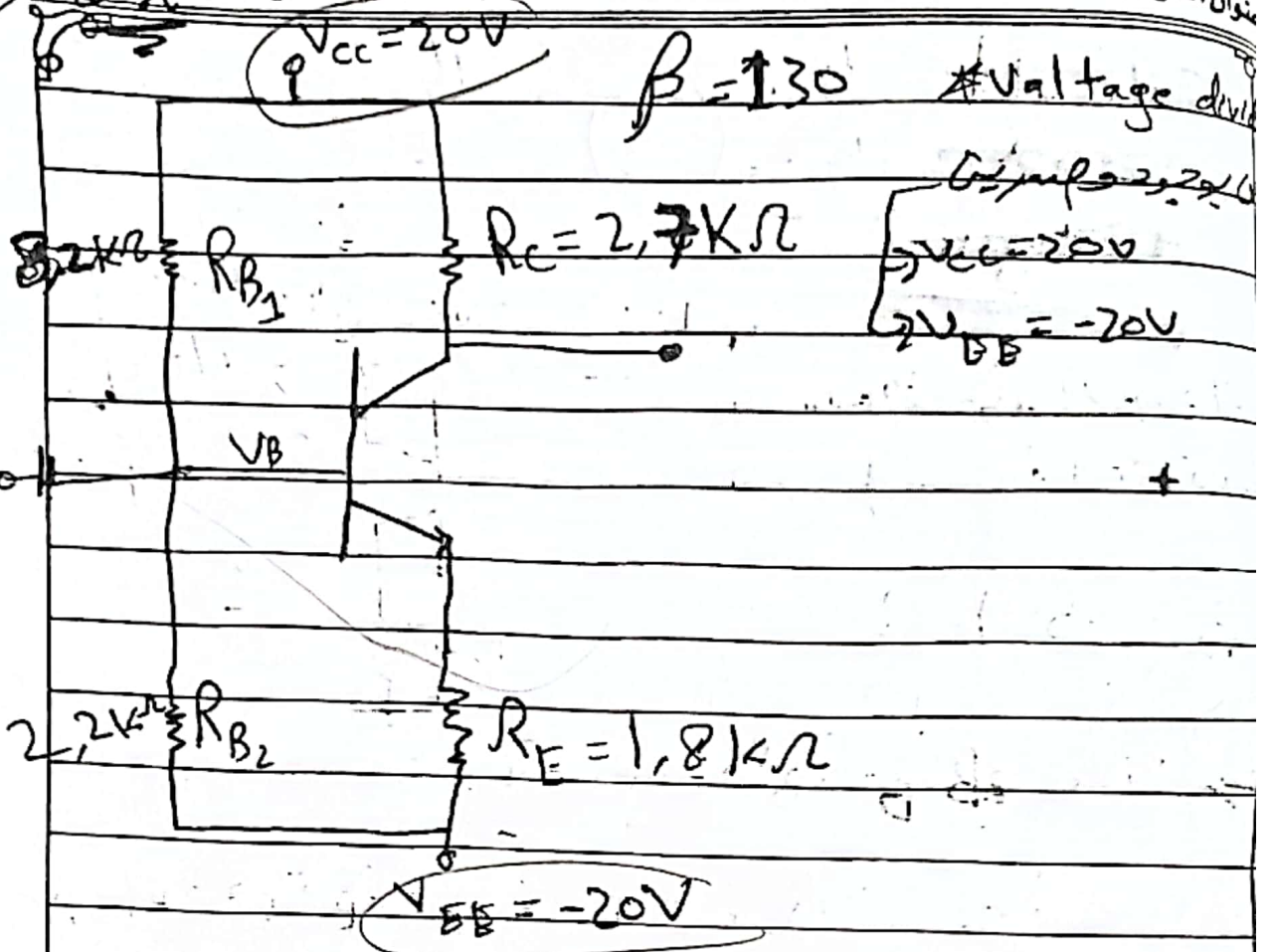
$$V_{CE} = V_C - V_E$$

$$V_{CC} - R_C I_C - V_C = 0$$

$$V_C = V_{CC} - R_C I_C$$

$$V_C = 10 - 3k \times 2.195 \times 10^{-3}$$

$$V_C = 3.415V$$



المطلوب: V_{BB} و R_{BB} (المطلوب)

$$R_{BB} = \frac{R_{B1} \times R_{B2}}{R_{B1} + R_{B2}} = \frac{8.2K \times 2.2K}{8.2K + 2.2K}$$

$$R_{BB} = 1.735K$$

$$V_{BB} = \frac{V_{CC} \times R_{B2}}{R_{B1} + R_{B2}} + \frac{V_{EE} \times R_{B1}}{R_{B1} + R_{B2}}$$

المقاومة

SUPER MAAS

المقاومة

Scanned with CamScanner

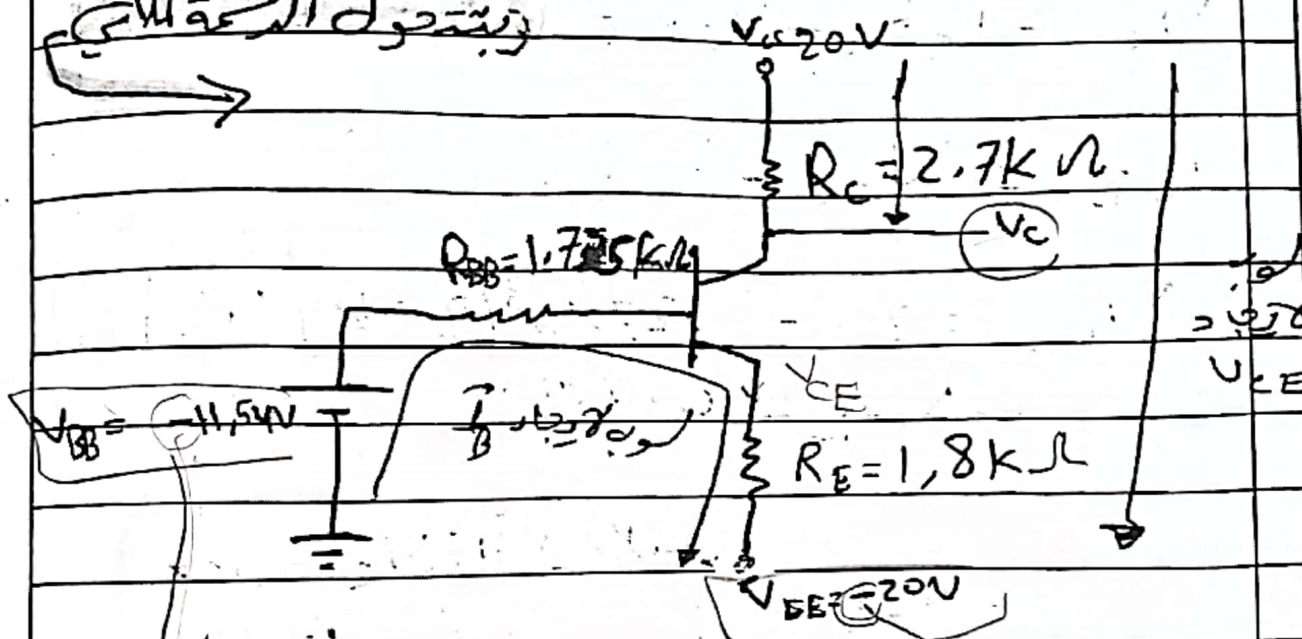
$$V_{BB} = \frac{20 \times 2.2k}{8.2k + 2.2k} + \frac{(-20) \times 8.2k}{8.2k + 2.2k}$$

$$V_{BB} = 4.23V \quad \text{---} \quad -15.77V$$

$$+ V_{BB} = 4.23 - 15.77 = -11.54V$$

$$\therefore V_{BB} = -11.54V$$

رسم دارة الترانزستور



$$-V_{BB} - I_B R_{BB} - V_{BE} - I_E R_E + V_{EE} = 0$$

$$I_B = \frac{V_{EE} - V_{BB} - V_{BE}}{R_{BB} + \beta R_E} = \frac{20 - 11.54 - 0.7}{1.735k + 130 \times 1.8k}$$

$$\therefore I_B = 32.91 \mu A$$

الواجب:

SUPER MAAS

Scanned with CamScanner

الممسوحة صوتياً بـ CamScanner

الممسوحة صوتياً بـ CamScanner

$$I_c = \beta I_B = 130 \times 32.91 \times 10^{-6}$$

$$I_c = 4.27 \text{ mA}$$

$$V_{CE} \Rightarrow \text{جهد المخرج}$$

$$V_{CC} - I_c R_c - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_c (R_c + R_E)$$

$$V_{CE} = 20 - 4.27 \times 10^{-3} (2.7 \text{ K} + 1.8 \text{ K})$$

$$\therefore V_{CE} = 0.785 \text{ V}$$

نوجد V_{CE} مثلاً له طريقين

طريق الأول

طريق الثاني

$$V_{CE} = V_C - V_E$$

$$0.785 = V_C - V_E$$

نوجد V_E من

$$V_E = I_E R_E$$

$$V_E = I_c \times R_E = 4.27 \text{ mA} \times 1.8 \text{ K}$$

$$V_E = 7.686 \text{ V}$$

$$0.785 = V_C - 7.686$$

$$V_{CC} - I_c R_c - V_C = 0$$

$$V_C = V_{CC} - I_c R_c$$

$$V_C = 20 - 4.27 \times 10^{-3} \times 2.7$$

$$V_C = 8.471 \text{ V}$$

$$V_{CE} = 0.785 + 7.686 \Rightarrow V_{CE} = 8.471 \text{ V}$$

by Eng: Youssif Yasser

المادة: 2019 (الزمن)

طوبى

R_{th} V_{th}

السؤال رقم 1

Calculate V_{CE} and I_C of fig. 1

$V_{CE} = ?$ $I_C = ?$

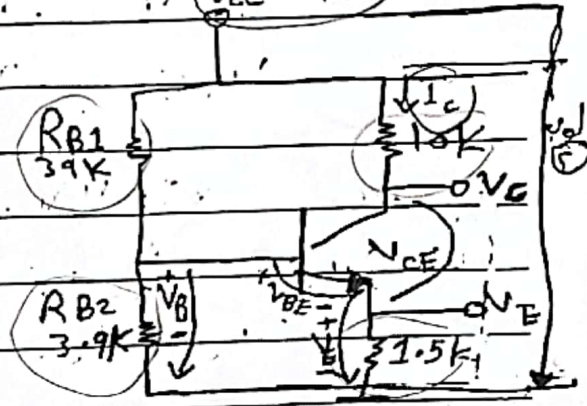
$V_{CC} = 22V$

Fig. 1

طوبى أو فكرة جديدة

عش فوجودة في

فنا اولي بيروج اوجد V_E و V_{CE} انما جديا لبارية
ال V_B عند ارجال الا قد قاتون فرق (التيهاتية نقطية)

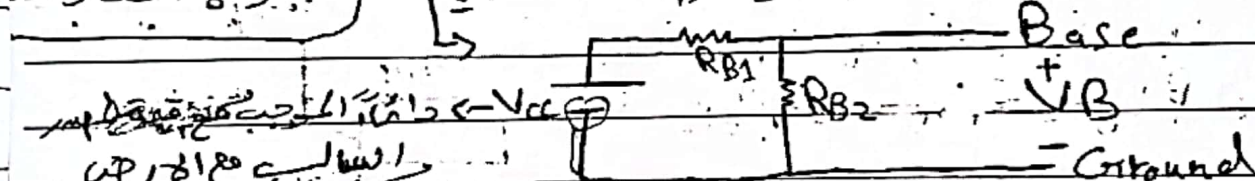


$V_B = \frac{V_{CC} \times R_{B2}}{R_{B1} + R_{B2}} = \frac{22 \times 3.9k}{39k + 3.9k} = 2V$

Voltage divider

فهرت القادة والدة
على V_B على تقسم
عبر على القادة

2. كية طلت القاسر وحيث ان مع ال V_B
يق القاسر ال Base وال V_{CE} Ground



$V_B = \frac{V_{CC} \times R_{B2}}{R_{B1} + R_{B2}}$

$V_B = 2V$

نما اوجدنا ال V_B نوصه ال V_E وذلك انما يعرف (التيهاتية نقطية)

$V_{BE} = V_B - V_E \Rightarrow V_E = V_B - V_{BE} = 2 - 0.7 = 1.3V$

$V_E = 1.3V$

التيهاتية نقطية I وبذلك استنتجنا ان $V_E = I_E R_E$

$V_E = I_E R_E \Rightarrow I_E = \frac{V_E}{R_E}$

$I_E = I_C$

$I_C \approx I_E = \frac{V_E}{R_E} = \frac{1.3V}{1.5k} = 0.867mA$

$I_C = I_E = 0.867mA$

مثال ١: حساب V_{CE} وذلك بأخذ I_C من جزء ١

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$V_{CE} = 22 - 0.867 \text{mA} (10\text{k} + 1.5\text{k}) = 12.029\text{V}$$

مثال ٢: حساب V_C وذلك بأخذ I_C من جزء ١

$$V_{CC} - I_C R_C - V_C = 0$$

$$V_C = V_{CC} - I_C R_C = 22 - 0.867 \times 10^{-3} \times 10 \times 10^3$$

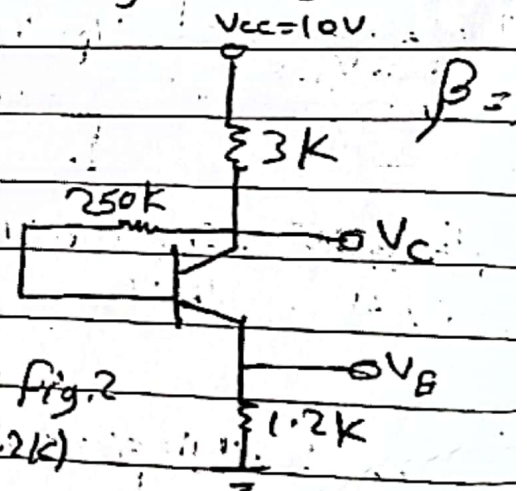
$$V_C = 13.33\text{V}$$

Q2: Calculate I_E and V_{CE} using voltage divider

نطبق نظام حل هذا النوع مرة أخرى

$$V_{CC} - I_C R_C - I_B R_B - V_{BE} - I_E R_E = 0$$

$$V_{CC} - \beta I_B R_C - I_B R_B - V_{BE} - \beta I_B R_E = 0$$



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{10 - 0.7}{250\text{k} + 50(3\text{k} + 1.2\text{k})}$$

$$I_B = 20.21 \mu\text{A}$$

$$I_C = \beta I_B = 50 \times 20.21 \times 10^{-6} = 1.010\text{mA}$$

نحسب I_E وذلك بأخذ I_C من جزء ٢

$$I_E = I_C + I_B = 1.010 \times 10^{-3} + 20.21 \times 10^{-6} = 1.030\text{mA}$$

$$V_{CE} = V_{CC} - R_C I_C - V_{CE} - R_E I_E = 0$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

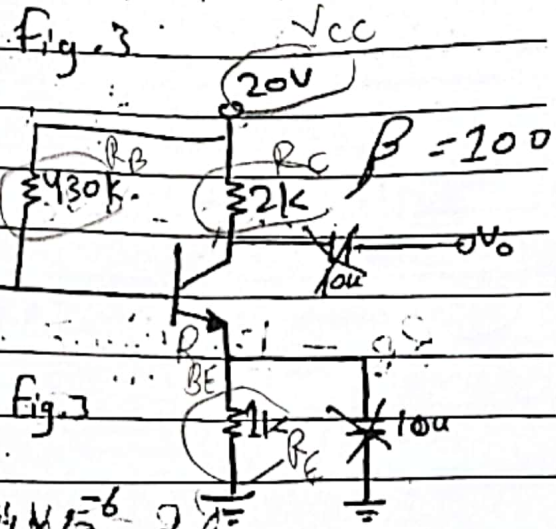
$$V_{CE} = 10 - 1.010 \times 10^{-3} (3\text{k} + 1.2\text{k})$$

$$V_{CE} = 5.76\text{V}$$

pl back

Q3/ Calculate V_{CE} and I_C of Fig. 3

① $I_B \Rightarrow V_{CC} - R_B I_B - V_{BE} - R_E \beta I_B = 0$



$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta R_E}$$

$$I_B = \frac{20 - 0.7}{430k + 100 \times 1k} = 36.41 \mu A$$

② $I_C = \beta I_B = 100 \times 36.41 \times 10^{-6} = 3.641 mA$

③ $V_{CE} \Rightarrow V_{CC} - R_C I_C - V_{CE} - R_E I_C = 0$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

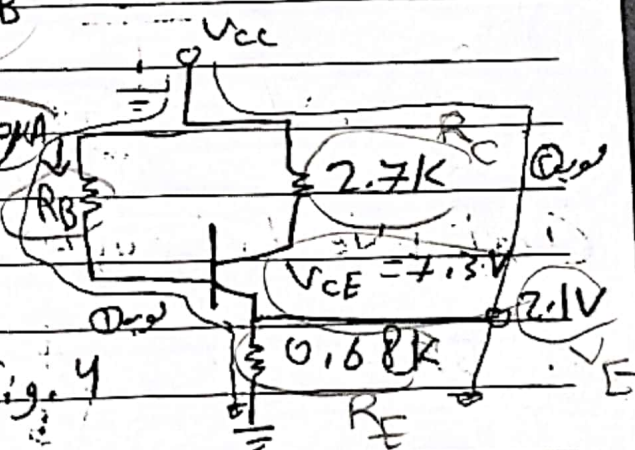
$$V_{CE} = 20 - 3.641 \times 10^{-3} (2k + 1k) = 9.077V$$

Q4/ Calculate β , V_{CC} , I_C , R_B

$I_B = 20 \mu A$
 $V_{CE} = 7.3V$ * $V_E = 2.1V$

① $\beta = \frac{I_C}{I_B}$

$I_C \approx I_E = \frac{V_E}{R_E} = \frac{2.1V}{0.68k} = 3.08 mA$



$$I_E \approx I_C = 3.08 mA$$

$$\beta = \frac{I_C}{I_B} = \frac{3.08 \times 10^{-3}}{20 \times 10^{-6}} = 154$$

for V_{CC} calculation

$$V_{CC} - R_C I_C - V_{CE} - R_E I_C = 0$$

$$V_{CC} = V_{CE} + I_C (R_C + R_E) = 7.3 + 3.08 \times 10^{-3} (2.7k + 0.68k)$$

$$V_{CC} = 14.71V$$

* $R_B \Rightarrow$ اختيار 10 ك

$$V_{CC} - R_B I_B - V_{BE} - (R_E I_E) = 0$$

$$V_E = 0.68 \times 16 \times 3.5$$

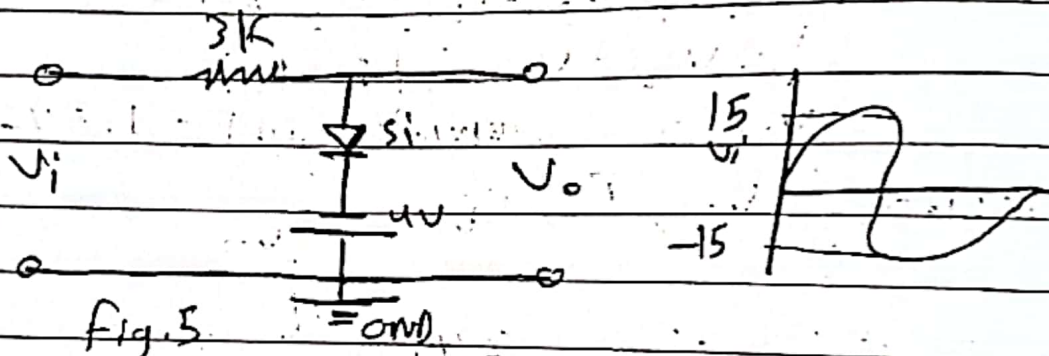
$$V_E = 2.0$$

$$V_{CC} - V_{BE} - V_E = R_B I_B$$

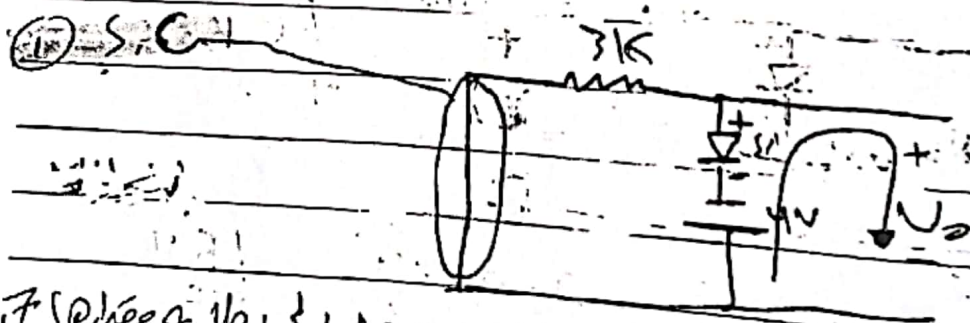
$$I_B = \frac{V_{CC} - V_{BE} - V_E}{R_B} = \frac{17.71 - 0.7 - 2}{20 \times 10^3}$$

$$R_B = 745.78 \text{ K}\Omega$$

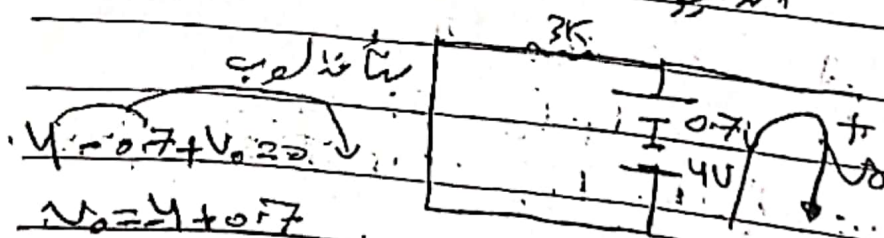
Q5 / Sketch and Calculate V_o of fig. 5



المطلوب من دارب دال V_o وكذا
نطبق في الدارة كما درست في قبل
نوع (Clipper Parallel)



الآن نحدد شكل V_o في الدارة



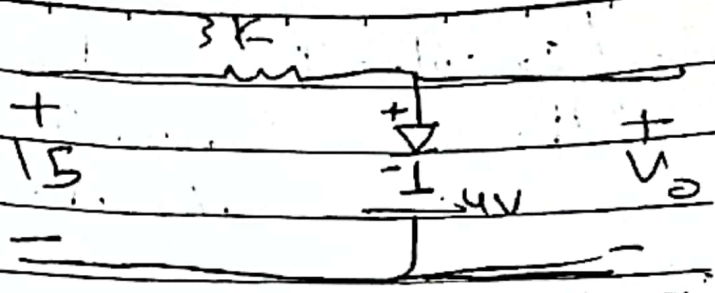
$$V_o = 0.7 + V_o = 2.0$$

$$V_o = 4 + 0.7$$

$$V_o = 3.3 \text{ Vdc}$$

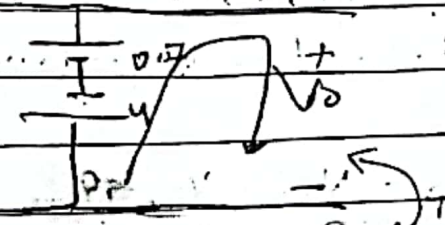
المطلوب من دارب دال V_o وكذا
نطبق في الدارة كما درست في قبل
نوع (Clipper Parallel)

② +ve half cycle



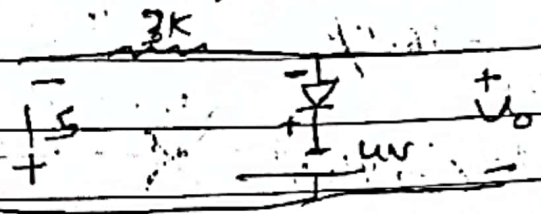
$$15 - 0.7 + V_o = 0$$

$$V_o = -14.3 \text{ Volts}$$



بالأولوية القوية من V_o والـ V_o

③ -ve half cycle

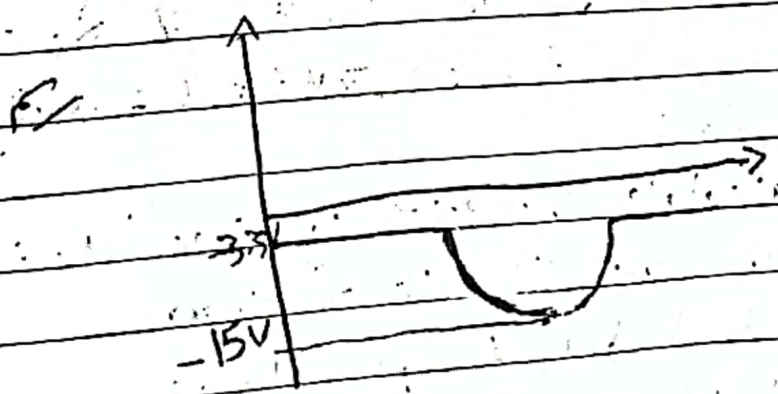
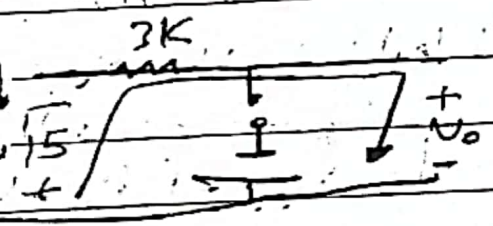


التيار في شغل بيكون open

$$V_R = I R \times R = 0 \times 3 = 0$$

$$15 - V_R + V_o = 0$$

$$V_o = -15 \text{ AC}$$



منه حلة في كتيبات الليبراري
منه حلة في كتيبات الليبراري
منه حلة في كتيبات الليبراري

$$R_{th} = \frac{39k \times 3.9k}{39k + 3.9k} = 3.545k\Omega$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + \beta R_E}$$

$$= \frac{2 - 0.7}{3.545k + 70 \times 1.5k} = 11.97 \mu A$$

$$I_C = \beta I_B = 70 \times 11.97 \times 10^{-6} = 0.837 mA$$

$V_{CE} \Rightarrow$ اذنيك لوب

$$V_{CE} = V_{CC} - I_C (R_E + R_C)$$

$$V_{CE} = 22 - 0.837 \times 10^{-3} (1.5k + 10k)$$

$$V_{CE} = 12.37V$$

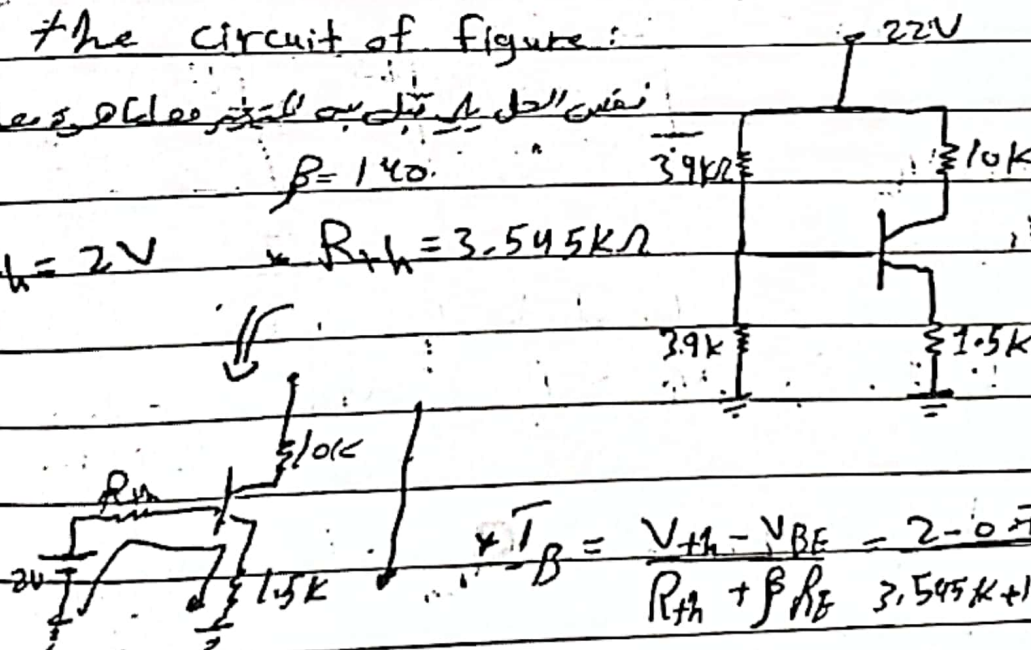
Q4/ Calculate the dc bias voltage V_{CE} and current I_C for the circuit of figure:

نفس الطر يقبل به لتخرج ملامح عامل ال

$$\beta = 140$$

$$V_{th} = 2V$$

$$R_{th} = 3.545k\Omega$$



$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + \beta R_E} = \frac{2 - 0.7}{3.545k + 140 \times 1.5k} = 6.08 \mu A$$

$$I_C = \beta I_B = 0.852 mA$$

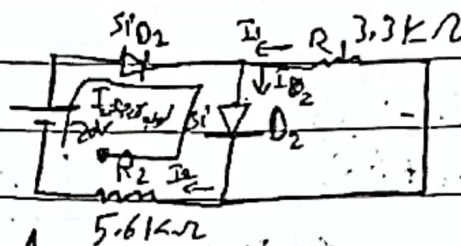
$$V_{CE} = 22 - 0.852 \times 10^{-3} (1.5k + 10k)$$

$$V_{CE} = 12.202V$$

Q.5/ Determine the current I_1 , I_2 and I_{D1} for the network of figure:

V_{D2} parallel with R_1

So $V_{D2} = V_{R1} = 0.7$

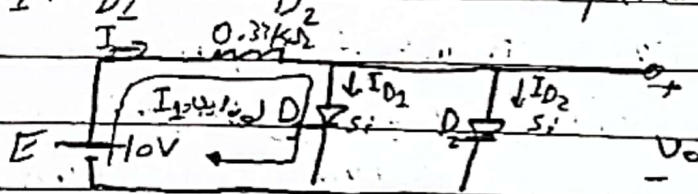


$$\therefore I_1 = \frac{V_{R1}}{R_1} = \frac{0.7}{3.3k} = 0.212 \text{ mA}$$

Loop 2: $E - V_{D1} - V_{D2} - I_2 R_2 = 0 \Rightarrow I_2 = \frac{E - V_{D1} - V_{D2}}{R_2} = \frac{20 - 0.7 - 0.7}{5.6k} = 3.32 \text{ mA}$

$I_1 + I_{D2} = I_2 \Rightarrow I_{D2} = I_2 - I_1 = 3.32 \text{ mA} - 0.212 \text{ mA} = 3.108 \text{ mA}$

Q.6/ Determine V_0 , I_1 , I_{D1} and I_{D2} for the parallel diode configuration of figure?



Q.1 V_0 parallel with V_{D1}

$V_0 = V_{D1} = 0.7$

Loop 1: $E - I_1 R_1 - V_{D1} = 0$

$$\Rightarrow I_1 = \frac{E - V_{D1}}{R_1} = \frac{10 - 0.7}{0.37k} = 28.18 \text{ mA}$$

③ $V_{D1} = V_{D2} \Rightarrow I_{D1} = I_{D2} = \frac{I_1}{2}$

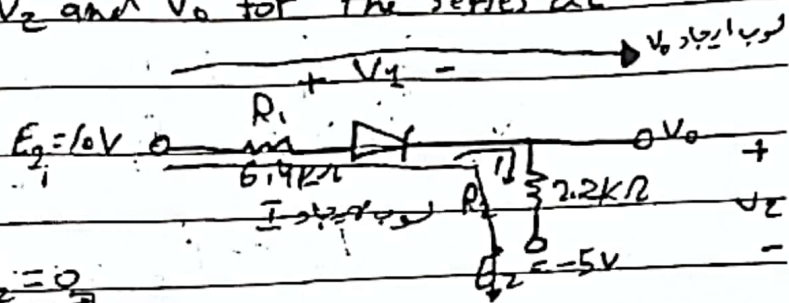
$$\Rightarrow \frac{I_1}{2} = \frac{28.18 \times 10^{-3}}{2} = 14.09 \text{ mA}$$

$\therefore I_{D1} = 14.09 \text{ mA}$

$\therefore I_{D2} = 14.09 \text{ mA}$

مكرر نفس السؤال في الامتحان / ارجو
تأكيد الجواب

Q2/ Determine I , V_1 , V_2 and V_0 for the series dc
configuration of Figure.



نوع I اتجاهه لوب

$$E_1 - R_1 I - V_0 - R_2 I + E_2 = 0$$

$$E_1 + E_2 - V_0 = R_1 I + R_2 I$$

$$I = \frac{E_1 + E_2 - V_0}{R_1 + R_2} = \frac{10 + 5 - 0}{6.4K + 2.2K} = \frac{15}{8600} = 1.66m$$

$$I = 1.66mA$$

$$V_1 = I \cdot R_1 = 1.66 \times 10^{-3} \times 6.4 \times 10^3 = 10.624V$$

$$V_2 = I \cdot R_2 = 1.66 \times 10^{-3} \times 2.2 \times 10^3 = 3.652V$$

$$E_1 - IR_1 - V_0 - V_2 = 0$$

$$V_0 = E_1 - (IR_1) - V_2 = E_1 + V_1 - V_0 = 10 - 10.624$$

$$V_0 = -1.324$$

Q3/ نفس السؤال في الامتحان في مادة
gate or and

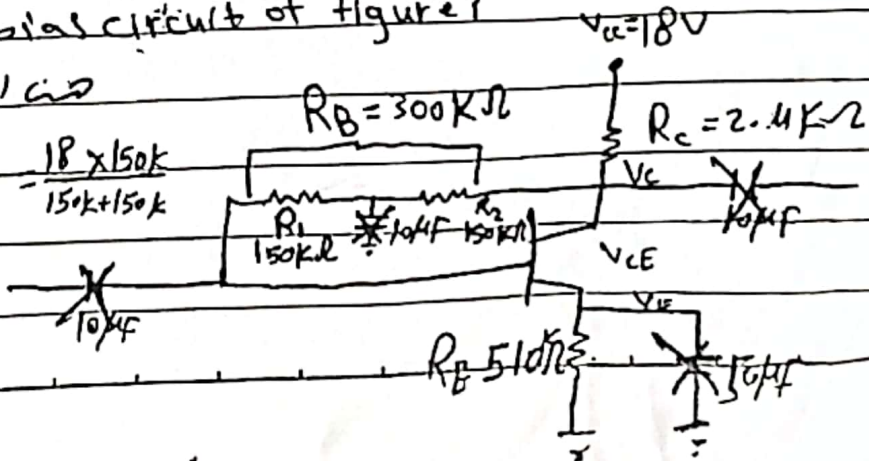
Q4/ نفس السؤال في الامتحان في مادة
gate or and

Q5/ Calculate the dc collector current I_c and voltage
 V_c for bias circuit of figure?

نوع β ليس مهم في

$$V_B = \frac{V_{CC} \times R_{B2}}{R_{B1} + R_{B2}} = \frac{18 \times 150k}{150k + 150k}$$

$$V_B = 9V$$



$$V_{BE} = V_B - V_E \quad \leftarrow V_E \text{ غير معروف}$$

$$0.7 = 9 - V_E \Rightarrow V_E = V_B - V_{BE} = 9 - 0.7$$

$$V_E = 8.3V$$

$$I_E \approx I_C \text{ لأن } I_E \text{ على شكل } I_C \text{ في حالة التوازن}$$

$$V_E = I_E R_E \Rightarrow I_E = \frac{V_E}{R_E} = \frac{8.3}{510k} = 0.016mA$$

$$I_B \approx I_C = \underline{0.016mA}$$

الآن نوجد V_C وذلك باستخدام $V_C = V_{CC} - I_C R_C$

$$\rightarrow V_{CC} - I_C R_C - V_C = 0$$

$$V_C = V_{CC} - I_C R_C = 18 - 0.016 \times 10^{-3} \times 2.4 \times 10^3$$

$$V_C = 18 - 0.016 \times 2.4 = \underline{17.96V}$$